

Problem Statement

VOIS AICTE Batch1 2026-2027 – Major Project: Seasonal Agriculture Performance Analysis

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Agricultural activities are influenced by seasonal variations in environmental conditions, farming practices, resource availability, and market conditions. As a result, agricultural performance may differ from one season to another.

However, raw agricultural data does not clearly explain how agricultural performance changes across seasons or what patterns can be observed in different seasonal conditions.

The problem is to analyze the given agricultural dataset and investigate seasonal differences in agricultural performance by identifying

meaningful patterns, trends, relationships, and variations within the available data.

Importance of the Problem Statement

Seasonal variations can influence different aspects of agricultural activities and performance. Understanding seasonal patterns through data analytics can help stakeholders:

- *Understand variations in agricultural performance*
- *Identify important seasonal trends*
- *Compare performance across different periods*
- *Understand changing environmental conditions*
- *Examine differences in resource usage*
- *Identify areas requiring further investigation*
- *Support evidence-based agricultural planning*

*For students, this problem provides an opportunity to work on a **focused real-world Data Analytics problem** rather than performing a generic analysis of the entire dataset.*

Dataset

The dataset contains farm-level agricultural records covering different seasons, locations, crops, environmental conditions, farming practices, production, costs, revenue, profit, and resource usage.

- **File:**

`seasonal_agriculture_performance_dataset.csv`

- **Records:** 4,000 farm entries

- **Coverage:** 8 Indian states, 8 crops, 4 irrigation methods, 3 seasons (Kharif, Rabi, Zaid)